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Review Article

The Functional Gastrointestinal Disorders in Infancy and the Fundamental Role of Probiotics: A Review on the Pathophysiology, Current Research and Future Therapy

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Abstract: Probiotics are the live microorganisms that they provide health benefits when consumed, generally by improving or restoring the gut flora. Now a day, the general public, researchers, International organization like WHO/FAO, pharmaceutical companies and food industries are becoming more interested in probiotics. According to the growth of, products, publications and public knowledge, research into the effectiveness of probiotics is gaining traction. "Let healthy microorganisms work for you in various domains acquire their advantages and take a break," says probiotics. Food digestion, the synthesis of helpful compounds to fight undesirable microorganisms, the complementing of the activities of missing digestive enzymes, the maintenance of the digestive system's pH, and so on are examples of such activity. Probiotics will boost the effectiveness of our digestive system's biological fermenters. Many writers have written on the history and development of probiotics as well as their many uses. In this study, we will primarily concentrate on three points: health improvement, infection control, and illness management, all of which might be avoided by using various forms of direct Probiotics or foods containing Probiotics. The most prevalent functional gastrointestinal diseases are infantile colic, constipation, functional abdominal pain, and irritable bowel syndrome. The present data on the use of probiotics in the treatment of this FGID will be reviewed in this chapter. Although the etiology of FGID is complex, the role of the gut microbiota in its development has been frequently stressed. As a result, the function of probiotics in their therapy is being investigated more closely. The use of Lactobacillus reuteri DSM 17938 at a dosage of 108 CFU/day for the treatment of infantile colic in breastfed babies currently has the greatest evidence of effectiveness. Lactobacillus rhamnosus at a dosage of 3 109 CFU and a multi-strain formulation for the treatment of IBS have limited but promising data. There is some evidence for the use of L. reuteri DSM 17938 at a dosage of at least 108 CFU/day in the treatment of FAP. Irritable bowel syndrome, functional dyspepsia, abdominal migraine, and functional abdominal pain not otherwise specified are among the pediatrics functional abdominal pain disorders, also known as diseases of gut-brain interaction, as described by the Rome IV diagnostic criteria. Functional abdominal pain problems affect 3-16 percent of the population, depending on the nation, age, and gender. The diagnosis is difficult, although it is based mostly on clinical symptoms and the elimination of alternative organic causes, with a focus on avoiding intrusive diagnostic techniques. Improved knowledge and treatment of these puzzling illnesses are expected in the next decades.

Keywords: Functional Abdominal Pain, Infantile Colic, Irritable Bowel Syndrome, Lactobacillus, Probiotics, Treatment's